



Generative Modeling by Neural Optimal Transport GAN NOT-GAN

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Main Goal of the Project

To bring a beautiful piece of theory to life:

- 1. understand what it takes to make it work
&**
- 2. understand how far it(we) can go.**

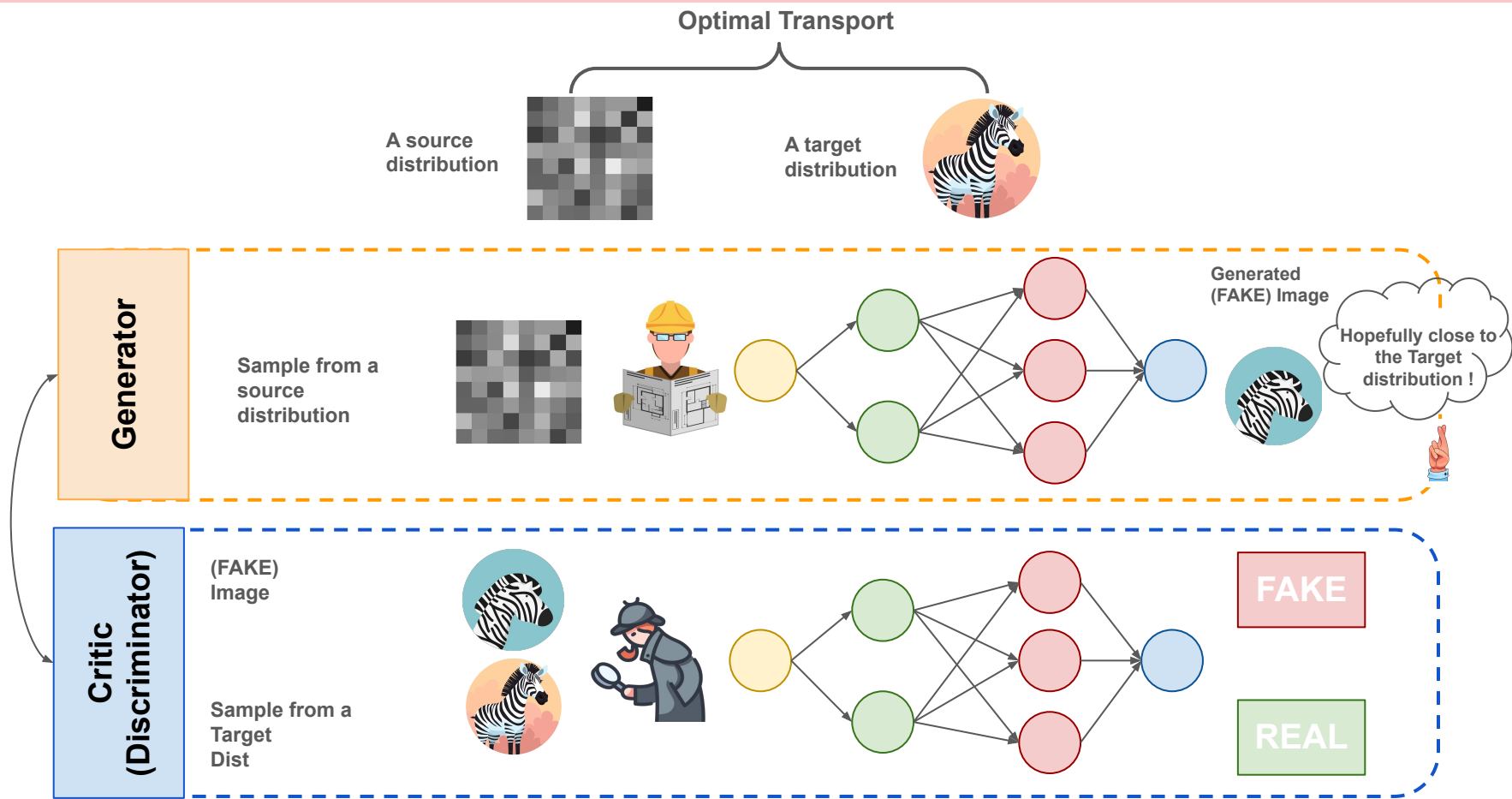
But First ...

What is a GAN

(and what does it have to do with
optimal transport)

?

GAN (Generative Adversarial Network)



Neural Optimal Transport GAN (NOT-GAN) vs Wasserstein WGAN (WGAN)

$\min_{\pi \geq 0} \max_{f \in \mathbb{R}^{\mathcal{Y}}} \mathcal{L}(\pi; f)$	NOT GAN	WGAN - GP
Critic Loss (Maximized)	$\mathcal{L}_D = f_\alpha(Y) - f_\alpha(\hat{Y}),$	$\mathcal{L}_D = f_\alpha(Y) - f_\alpha(\hat{Y}) - \lambda \mathbb{E}_{\tilde{Y}} \left[\left(\ \nabla_{\tilde{Y}} f_\alpha(\tilde{Y})\ _2 - 1 \right)^2 \right]$
Generator Loss (Minimized)	$\mathcal{L}_G = c(X, \hat{Y}) - f_\alpha(\hat{Y}),$	$\mathcal{L}_G = -f_\alpha(\hat{Y})$
Objective	Minimizes W2 (L2 squared)	Approximates W1 (L1)
Critic	Unconstrained	Constrained via gradient penalty (soft Lipschitz)
Transport Cost	Explicit L2 cost (generator penalized for inefficient paths)	No explicit cost term (does not penalize how far the generator moves mass)

Optimization Strategies

01

SGD

$$\theta_t = \theta_{t-1} - \eta g_t$$

02

Adam

$$\theta_t = \theta_{t-1} - \eta u_t, \quad u_t = \frac{\text{smoothed gradient}}{\text{smoothed magnitude}}$$

03

Optimism

$$\theta_t = \theta_{t-1} - \eta(2u_t - u_{t-1}) = \theta_{t-1} - \eta \underbrace{[u_t + (u_t - u_{t-1})]}_{\text{predicted change}}$$

04

Anchoring

$$\theta_t = \alpha_t \theta_{\text{anchor}} + (1 - \alpha_t) \theta_t, \quad \alpha_t = \frac{1}{t + 1}$$

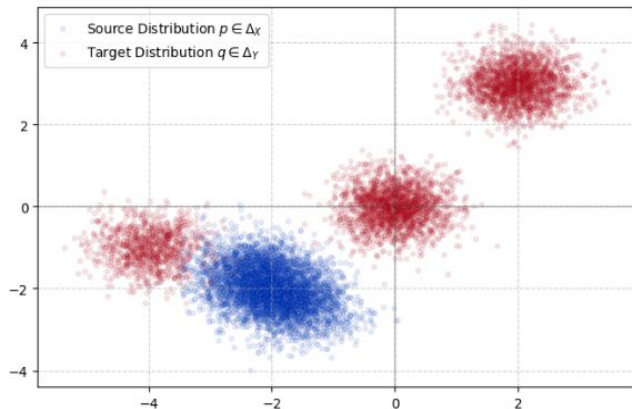
From theory to practice ...

Does it work?

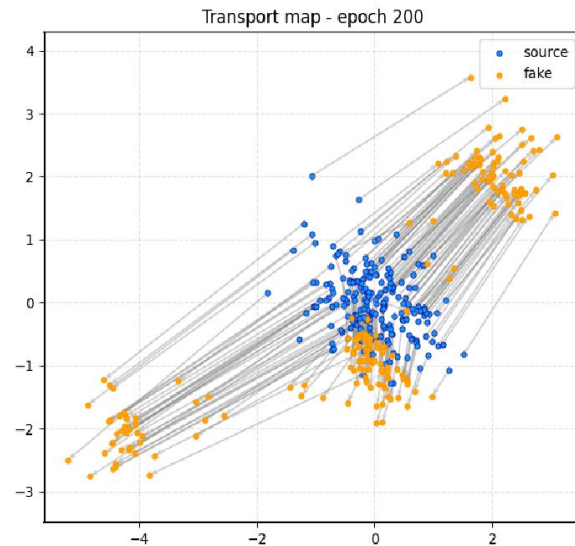
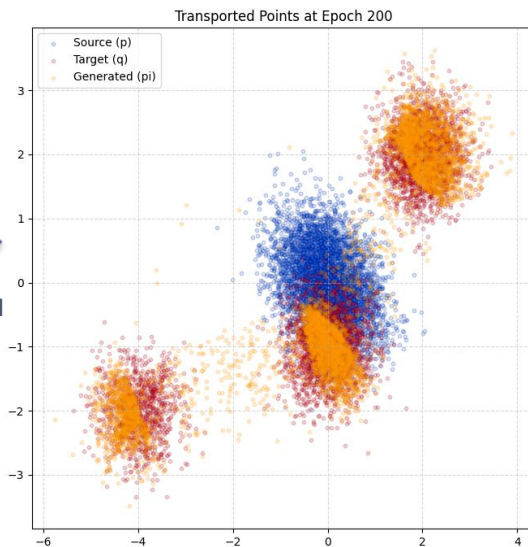
Proof of Concept: 2D Experiments

2D 1 digit 10 digits

Source & Target Distribution



NOT-GAN
transport



Base Hyperparameters

Reparam. noise σ

0.01

Critic LR (γ)

1×10^{-4}

Gen LR (η)

1×10^{-4}

Samples

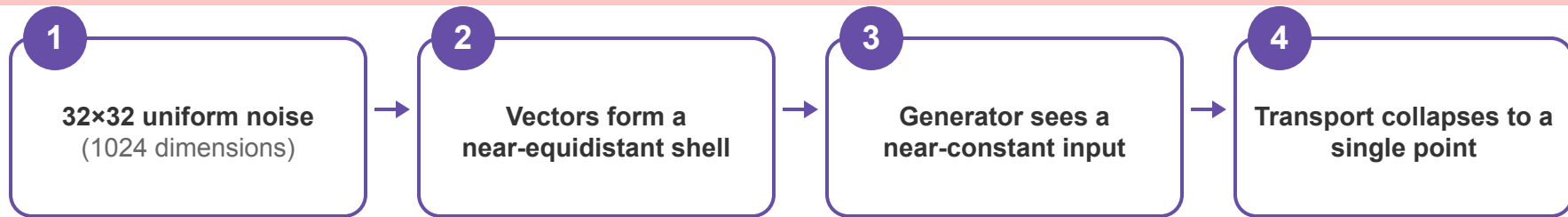
100,000

Optimizer

**Anchored
Optimistic Adam**

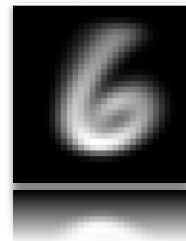
High Dimensional Mode Collapse

2D 1 digit 10 digits



Generated (any noise input)

=



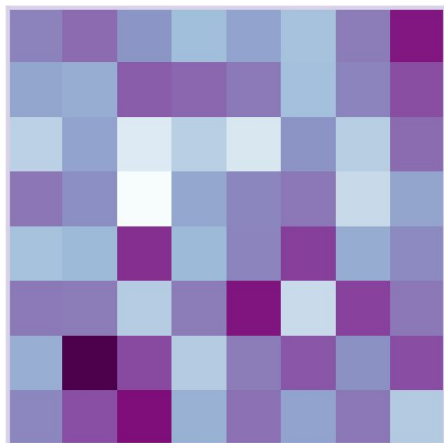
Pixel-wise mean of 6,000 sixes

High-dimensional noise washes out all representational variance, so the network learns a constant transport map equal to the target mean.

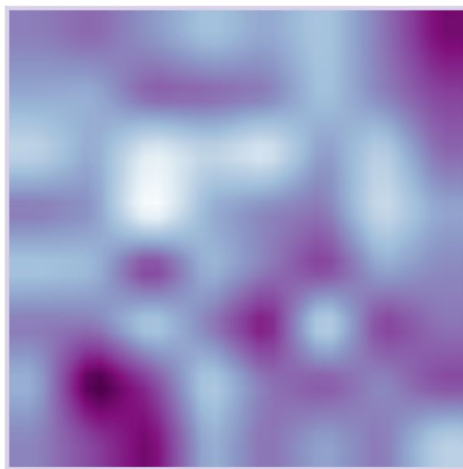
Low-Frequency Noise

Sample 8×8, bilinearly upsample to 32×32

Distinct blobby inputs → sharp, diverse digits



Drawn Coarse
8×8 Input Grid



Bilinear Upsample
32×32 Smooth Blob



Diverse 6's Return
High-fidelity digit synthesis

FID

Do the two distributions match overall?

Feature space similarity via MNIST-tuned LeNet (architecture to extract features)

Precision · Density

Do the fakes look real?

Fidelity (calculated on a trained classifier)

Recall · Coverage

Did we cover the full variety of real data?

Diversity (calculated on a trained classifier)

Pixel- W_2

What is the raw transport cost?

Exact squared euclidean transport cost in raw image space

Taming the Critic

2D 1 digit 10 digits

Adam

Explodes past 10^9 and model dies

Optimistic Adam

Damps the oscillation, but drifts

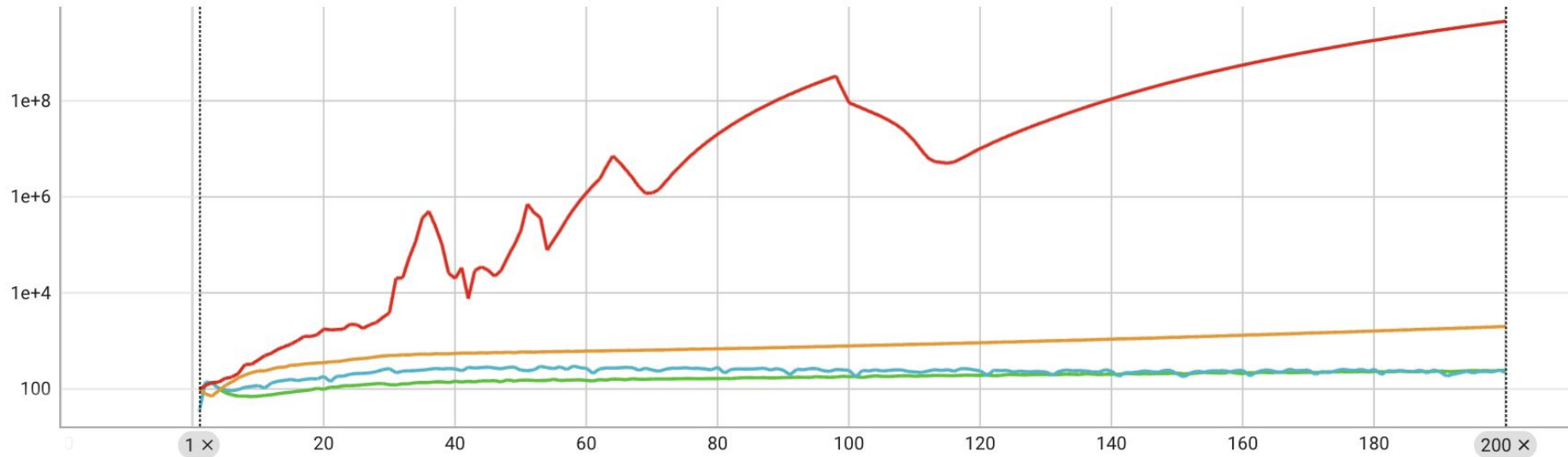
Anchored Adam + Reset

Stops the drift, too rigid alone

Anchored Optimistic Adam + Reset

Bounded ≈ 250 · FID 32.4

Critic Gradient Norm

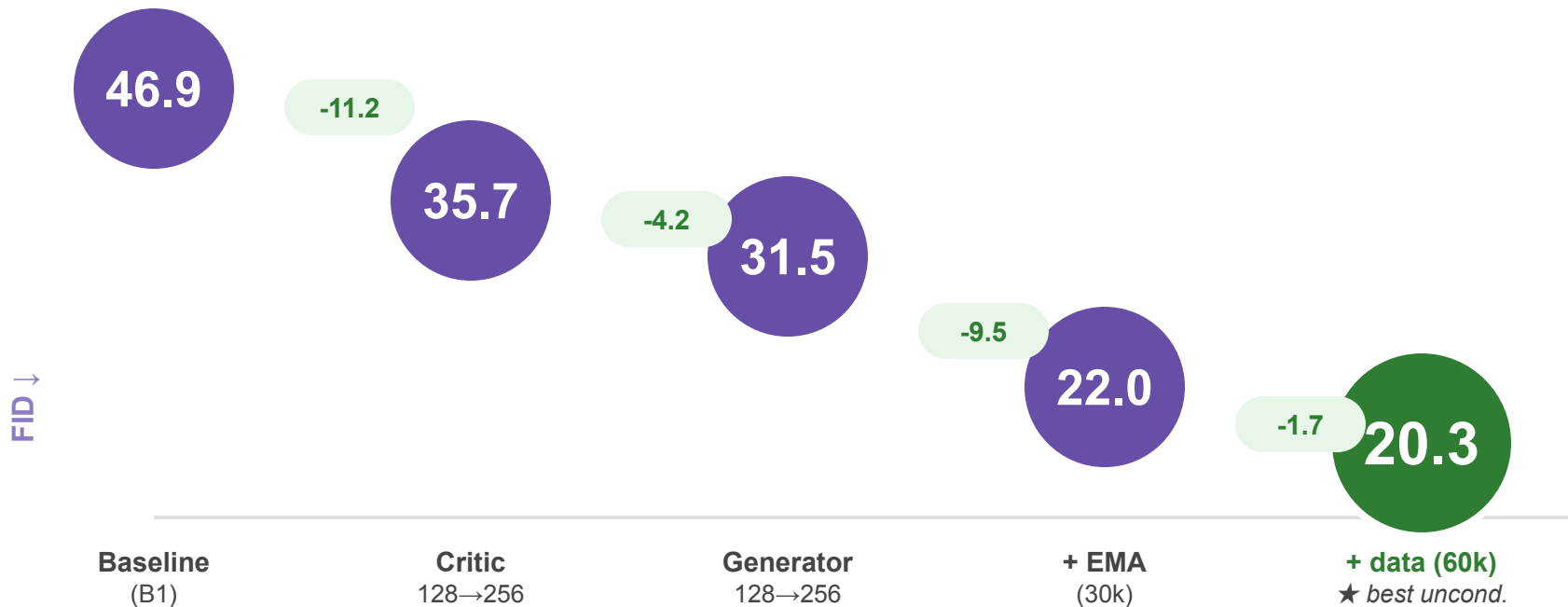


Critic gradient norm (log scale)

Ablation Ladder

2D 1 digit 10 digits

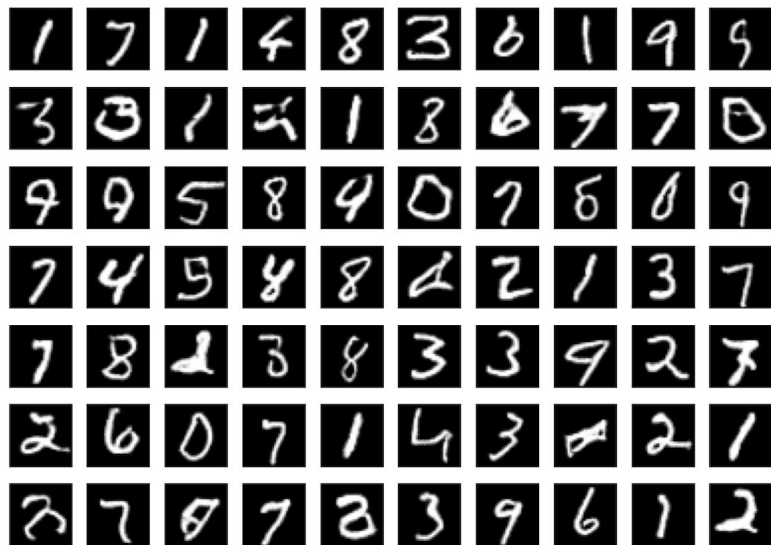
Each model inherits the best settings of the previous. FID scores show progressive improvement.



Unconditional Vs Conditional Generation

Unconditional (B10)

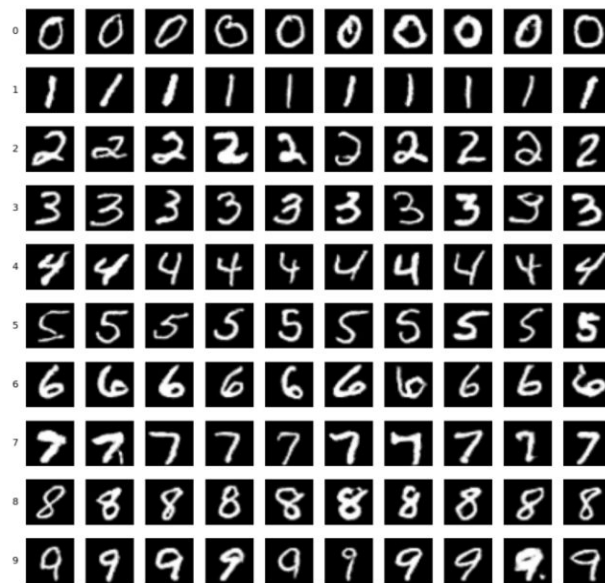
100 random samples (mean conf 0.94)



Random digits distributed across all classes

Conditional (C2)

10 samples per class (mean conf 0.99)

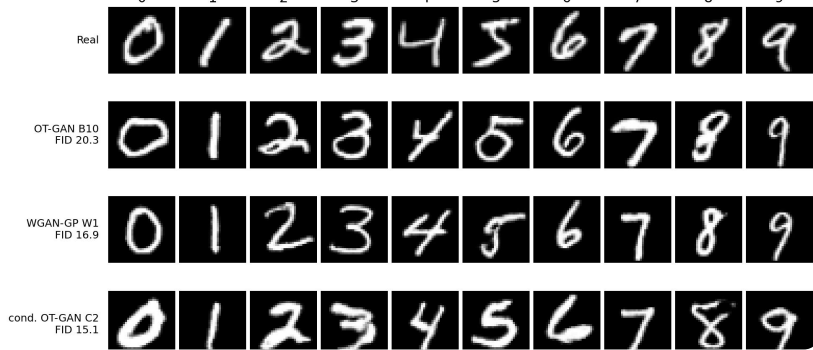


Rows = requested digit; precise class control

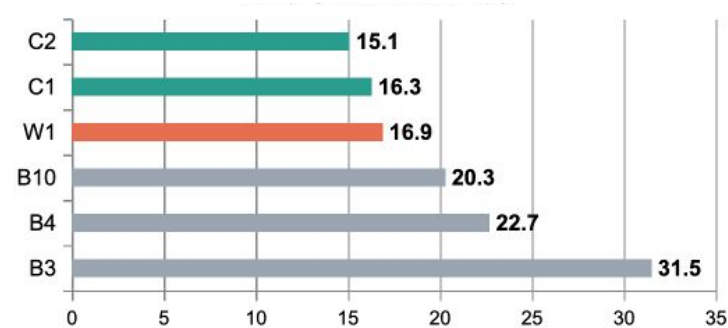
MNIST Models Comparison Performance



Sample Visual Comparison



FID Score Analysis (Lower is Better)

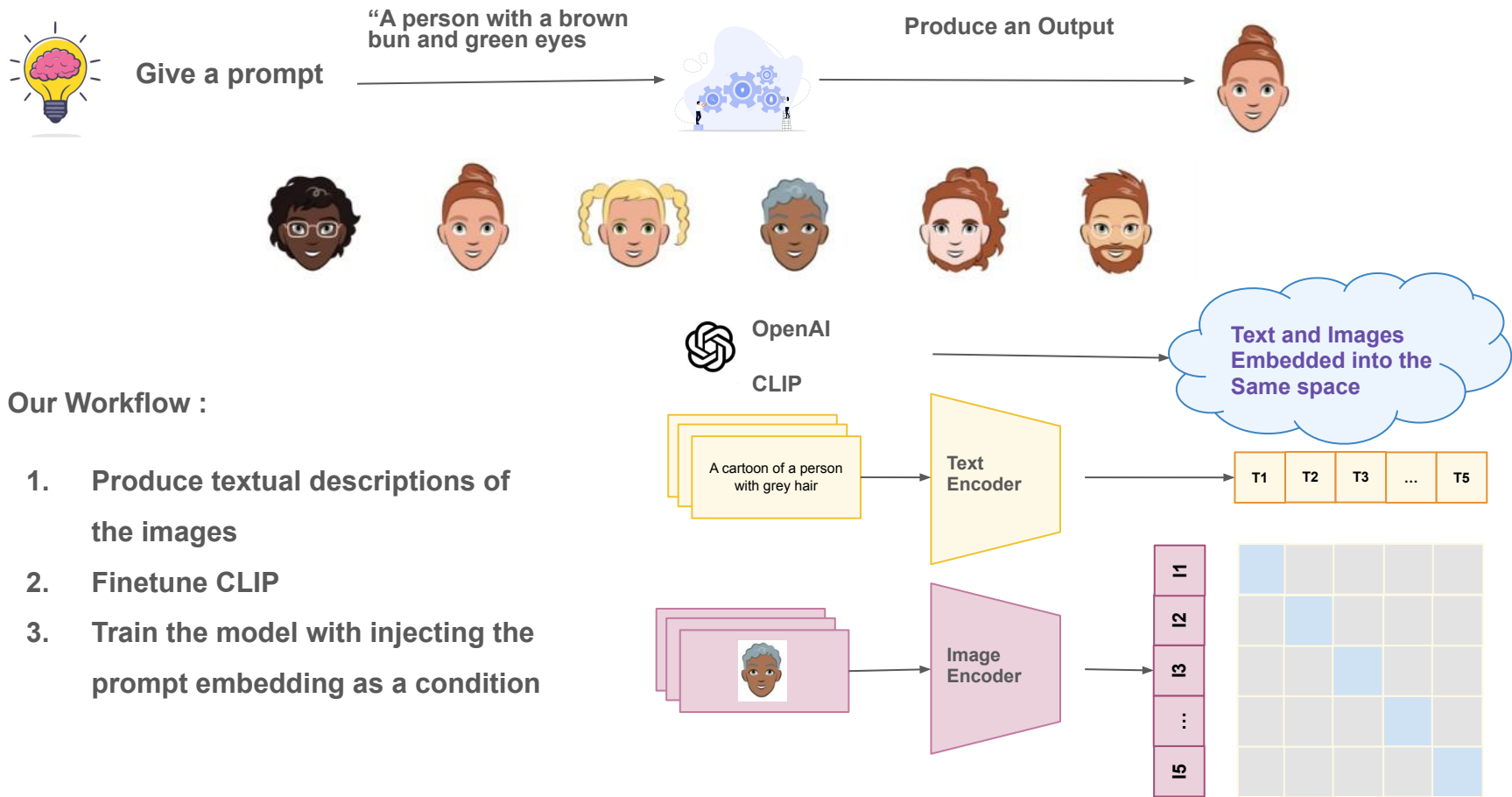


Detailed Evaluation Metrics

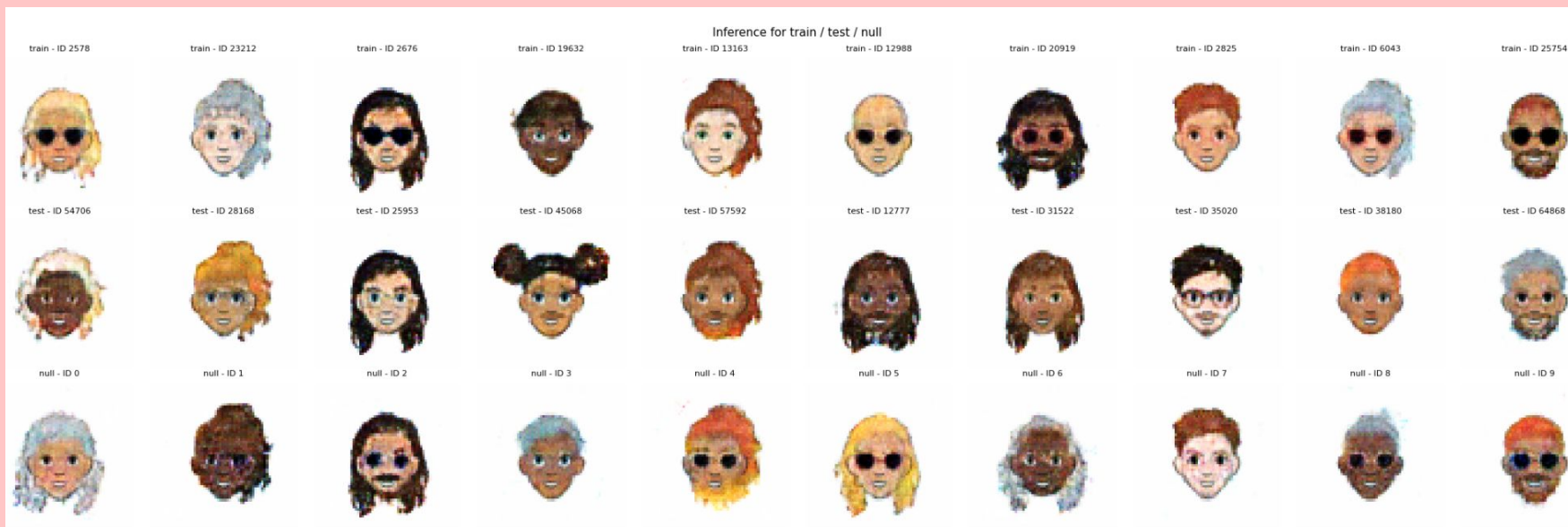
Model	Train	pixel-W ₂ ↓	Coverage ↑	Density ↑	Mem. (NN) ≈1
C1 (ours)	30k	45.3	0.96	0.96	0.96
C2 (ours)	60k	46.1	0.94	0.88	0.97
W1	30k	45.3	0.94	0.87	0.97
B10	60k	46.1	0.88	0.68	1.00
B4	60k	48.0	0.84	0.59	0.99
B3	30k	50.7	0.76	0.46	0.97

Bold = best in column. Mem. (NN) = nearest-neighbour distance to train ÷ to held-out test; ≈1.0 ⇒ no data-copying. FID shown in chart ↑

Text-to-Image Synthesis

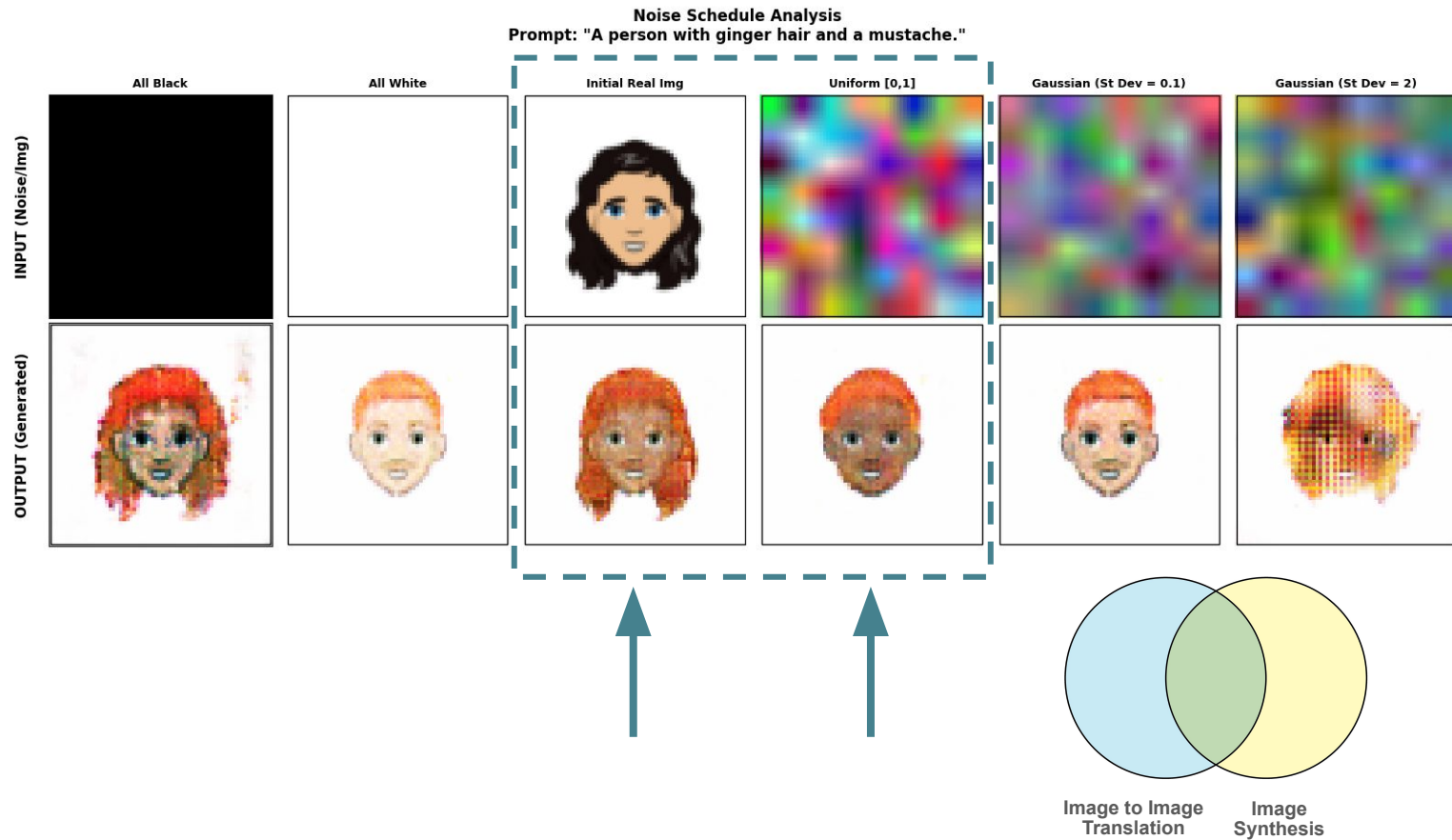


Results



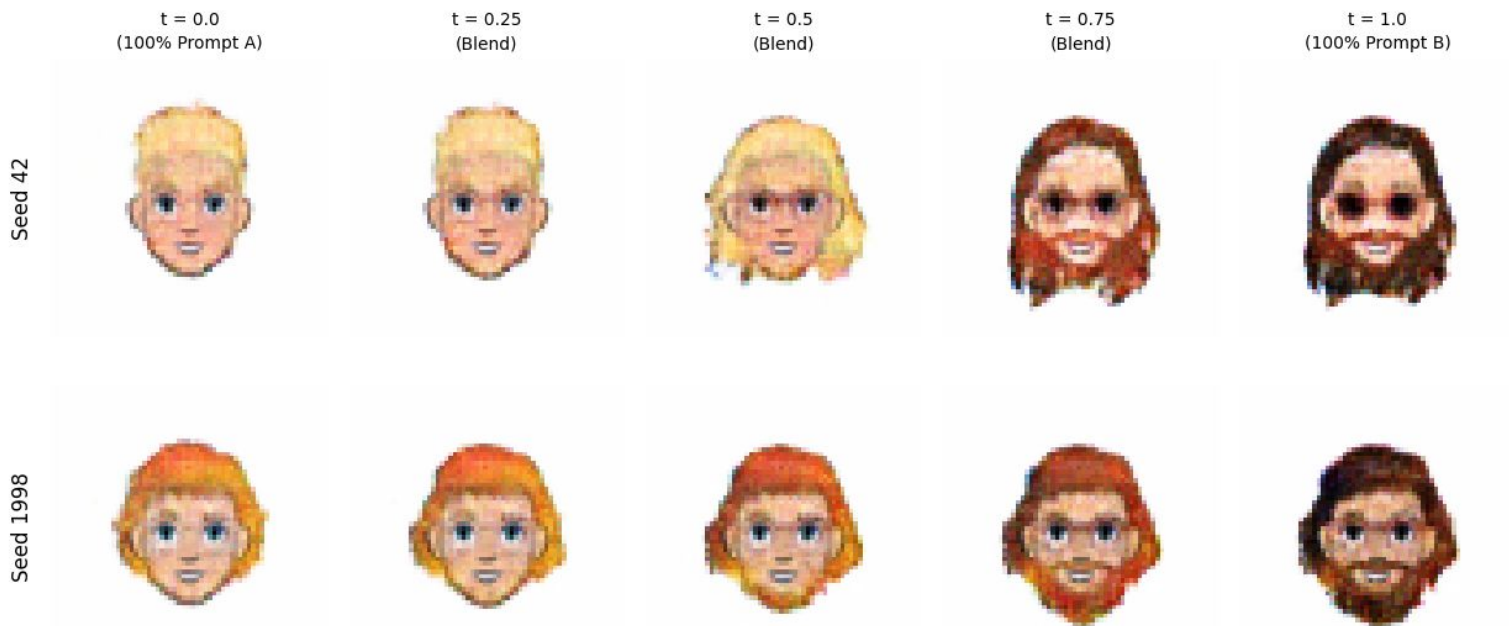
1. Diverse sample
2. Good Test Images
3. Good images when no prompt is given

Results



Results

Latent Space Interpolation with SLERP
Prompt A: 'A person with short blonde hair and blue eyes'
Prompt B: 'A person with long black hair, a beard, and red glasses'



Lessons Learned (& Turning Points)

01

Figuring out the optimization strategy

02

Scaling the asymmetric update schedule

03

Solving the high-frequency noise failure

04

Figuring out the right architecture!

05

Writing code for custom parameter update functions & training loop

06

Compute Constraint!

Thank You !

